

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

A museum rents tablets to visitors. The museum earns revenue of **\$14** for each tablet rented for the day. On Wednesday, the museum earned **\$406** in profit from renting tablets after paying daily expenses of **\$112**. How many tablets did the museum rent on Wednesday?
(**profit = total revenue – total expenses**)

Correct Answer: 37

Rationale

The correct answer is **37**. It's given that the museum earns revenue of **\$14** for each tablet rented for the day. It's also given that on Wednesday, the museum earned **\$406** in profit from renting tablets after paying daily expenses of **\$112**. Let ***x*** represent the number of tablets the museum rented on Wednesday. It follows that the total revenue can be represented by the expression **14*x***. Because **profit = total revenue – total expenses**, the equation **406 = 14*x* – 112** represents this situation. Adding **112** to both sides of this equation yields **14*x* = 518**. Dividing both sides of this equation by **14** yields ***x* = 37**. Therefore, the museum rented **37** tablets on Wednesday.